Liver Function and Oxidative Stress in Adult Male Wistar Rats Fed High Fat Diet

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ABSTRACT

This study investigated the effects of high fat diet formulated from lard on some serum liver function parameters and oxidative stress in adult male Wistar rats. Twenty-four rats were divided into four groups; a control and three experimental groups fed diets containing 5%, 10%, or 25% lard-based diets for eight weeks. Blood sample was collected thereafter for the estimation of serum Aspartate Aminotransferase (AST), Alanine Aminotransferase (ALT), Alkaline Phosphatase (ALP) activitives and liver tissue lipid peroxidation (Malondialdehyde - MDA). The data were analsed for statistical significance at $p < 0.05$ and presented in table. The results showed that fat diet caused a dose-dependent increase in the activities of serum liver enzymes and MDA level and were statistically elevated in the 10% and 25% fed groups compared with the control. These findings suggest that consumption of a high-fat diets induced significant hepatocellular damage and oxidative stress, with the severity directly linked to the dietary fat contents.

Keywords: High fat diet, Lard, Liver enzymes, Lipid peroxidation, Rats.

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1. INTRODUCTION

The liver is a vital metabolic organ that processes nutrients and detoxifies the body, making its health important for overall physiological balance. The modern diet, which often includes the frequent consumption of high-fat diets (HFD), presents a challenge to hepatic well-being. This dietary pattern, rich in saturated and trans fats, is a known precursor to non-alcoholic fatty liver disease (NAFLD), a condition that begins with simple steatosis and can advance to steatohepatitis, fibrosis, and cirrhosis [1]. An assessment of liver functions after HFD consumption is necessary as an investigation into a mechanism of diet-induced metabolic disease. The relationship is complex and involves metabolic disturbances that require exploration.

A key concept in this process is insulin resistance, a condition where cells do not respond properly to insulin, leading to hyperinsulinemia and an increased flow of free fatty acids to the liver, which promotes triglyceride accumulation and hepatic lipid overload [2]. This metabolic state fosters oxidative stress, where an imbalance between reactive oxygen species and antioxidants results in cellular damage, inflammation, and further progression of liver injury [3, 4].

A careful assessment of liver function can act as an early detection system [5], hence, by linking specific dietary fat profiles with measurable functional changes, this research can help shape nutritional guidance and public health policies focused on prevention. Furthermore, finding early biomarkers of HFD-induced liver dysfunction could change screening methods, allowing for action before irreversible damage occurs [6]. This study is therefore carried out to assess some liver functions enzymes and liver tissue oxidative stress in adult male wistar rats fed with high fat diet formulated from lard.

2. MATERIALS AND METHODS

Experimental animals: A total of twenty-four (24) adult male Wistar rats, with an initial weight range of 150-180g, were procured for this study. The animals were housed in standard polypropylene cages in the Physiology Laboratory, Ambrose Alli University, Ekpoma, under controlled conditions of temperature (22±2°C) and a 12-hour light/dark cycle. All rats were allowed free access to water and were acclimatized to the laboratory environment for a period of one week prior to the commencement of the experimental procedures. The study protocol was reviewed and approved by the institutional animal ethics committee, and all experiments were conducted in strict adherence to the guidelines for the care and use of laboratory animals.

Animals grouping: Following the acclimatization period, the rats were randomly divided into four (4) dietary groups of six (6) animals each. Group 1 served as the control and was fed a standard rat chow (100g) daily. Group 2, 3 and 4 served as the low-, mid-, and high-fat diet groups and received 95g rat chow + 5g lard, 90g rat chow + 10g lard, and 75g rat chow + 25g lard respectively. All groups had *ad libitum* access to clean drinking water throughout the entire experimental duration.

Substance preparation: The experimental diets were prepared daily to ensure freshness and prevent rancidity. Briefly, The specified amounts of lard (5g, 10g, and 25g, respectively) were melted at a low temperature and thoroughly mixed with the rat chow to ensure a homogenous blend. The control diet consisted of standard rat chow.

Substance administration: The prepared diets were *ad libitum* administered to the respective groups daily for a period of eight (8) weeks. The daily feed ration was provided at a fixed time each morning, and any leftover feed from the previous day was removed.

Sample collection: At the end of the eight-week feeding period, the animals were fasted overnight but allowed access to water. The rats were then humanely euthanized under mild anaesthesia. Blood samples were collected via cardiac puncture into plain sample bottles. The blood was allowed to clot and then centrifuged at 3000 rpm for 10 minutes to separate the serum. The parameters assessed to evaluate liver function and oxidative stress included the serum enzymes Aspartate Aminotransferase (AST), Alanine Aminotransferase (ALT), and Alkaline Phosphatase (ALP).

Subsequently, the animals were dissected, and the liver from each rat was carefully excised, weighed, and a portion was homogenized in an ice-cold phosphate buffer for the assay of lipid peroxidation marker Malondialdehyde (MDA).

Statistical analysis: The data were subjected to a one-way analysis of variance (ANOVA) to test for significant differences among the group means. A p-value of less than 0.05 ($p < 0.05$) was considered to be statistically significant. All data were expressed as Mean ± Standard Error of the Mean (SEM).

3. RESULTS

The results showed the effect of varying levels of dietary lard on liver enzymes and liver tissue oxidative stress (Table 1). High-fat diets induced a dose-dependent increase in liver function enzymes and liver tissue MDA. Compared to the control, the 10% and 25% fat diets presented significant increase in the levels of AST (45.32 ± 4.15 U/L vs 68.45 ± 6.37 U/L vs 95.71 ± 8.94 U/L), ALT (38.67 ± 3.82U/L vs 59.83 ± 5.72 U/L vs 88.46 ± 8.11 U/L), ALP (125.50 ± 10.33U/L vs 167.20 ± 15.44 U/L vs 245.83 ± 22.18 U/L).

Similarly, liver tissue MDA was significantly increased in the 10% (3.02 ± 0.41 nmol/mg tissue) and 25% fat diet groups (4.57 ± 0.62 nmol/mg tissue) compared to the control (1.85 ± 0.21m nmol/mg tissue).

Table 1: Effect of varying levels of dietary lard on liver function enzymes and oxidative stress marker in Wistar rats.

Group	AST (U/L)	ALT (U/L)	ALP (U/L)	Liver tissue MDA (nmol/mg tissue)
Control	45.32 ± 4.15	38.67 ± 3.82	125.50 ± 10.33	1.85 ± 0.21
Low Fat (5%)	52.18 ± 5.01	44.91 ± 4.55	138.75 ± 12.67	2.14 ± 0.28
Mid Fat (10%)	68.45 ± 6.37*	59.83 ± 5.72*	167.20 ± 15.44*	3.02 ± 0.41*
High Fat (25%)	95.71 ± 8.94*	88.46 ± 8.11*	245.83 ± 22.18*	4.57 ± 0.62*

Values are expressed as Mean ± Standard Deviation. *significant difference compared with control.

4. DISCUSSION

This study provides evidence on the dose-dependent hepatotoxic effects of a lard-based high-fat diet in a rodent model, further showing the significant and often underestimated impact of even moderately elevated fat consumption on liver integrity. The escalating prevalence of non-alcoholic fatty liver disease (NAFLD) globally necessitates a deeper understanding of how specific dietary components, like saturated fats, initiate and propagate liver injury [7].

The significant elevation in serum AST, ALT, and ALP levels in the mid and high-fat groups compared to the control indicates substantial hepatocellular damage and cholestasis (Table 1). These findings align with the work of Li et al. [8], who reported a marked increase in liver enzymes in Sprague-Dawley rats fed a 45% fat diet for 12 weeks. Similarly, a clinical study by Yoo et al. [9] found a positive association between saturated fat intake and elevated ALT levels in human subjects with NAFLD. The parallel and significant rise in hepatic MDA, a reliable marker of lipid peroxidation, in our high-fat groups is consistent with results from a study by Pandey et al. [10], which linked HFD-induced oxidative stress to mitochondrial dysfunction in hepatocytes. This concordance with existing literature strengthens the premise that dietary fat overload directly inflicts liver injury.

The physiological explanation for these observations lies in the metabolic processing of excess free fatty acids in the liver, which promotes triglyceride accumulation and subsequently induces endoplasmic reticulum stress. This stress response disrupts normal cellular function, leading to the leakage of enzymes like AST and ALT from hepatocytes into the bloodstream, while simultaneously generating reactive oxygen species that peroxidize membrane lipids, producing MDA [11].

When comparing the experimental groups, the graded, dose-response relationship observed where the high-fat diet group exhibited significantly worse outcomes than the mid-fat group, which in turn was worse than the low-fat group (Table 1). This pattern corroborates the findings of a meta-analysis by Garcia et al. [12], which concluded that the risk for progressive liver fibrosis in NAFLD patients increases linearly with the degree of transaminase elevation. However, our results appear to contrast with a study by Silva et al. [13], which found no significant difference in liver damage between mice fed 20% and 40% fat diets, suggesting a possible species-specific or diet-composition threshold.

The significant difference in ALP between the mid and high-fat groups points towards progressive biliary involvement at higher fat concentrations, a finding supported by recent

research from Chen et al. [14] that linked severe steatosis to impaired bile acid metabolism. The physiological basis for this dose-dependent effect is that as the dietary fat load increases, the liver's capacity for beta-oxidation and lipid export via VLDL becomes saturated. This exacerbates hepatic steatosis, amplifies oxidative stress through greater mitochondrial and peroxisomal fatty acid oxidation, and triggers a more robust inflammatory response via Kupffer cell activation, thereby accelerating the transition from simple steatosis to steatohepatitis [15].

The significant rise in MDA levels across the groups, particularly in the High Fat (25%) group, further shows the role of oxidative stress in the pathogenesis of high-fat diet-induced liver damage, with the escalating enzyme (AST, ALT, ALP) providing clear evidence of progressive hepatocellular injury and cholestasis

CONCLUSION

In conclusion, the findings from this study demonstrate that consumption of a lard-based high-fat diet induces significant, dose-dependent liver injury and oxidative stress in Wistar rats. These insights reinforce the role of dietary fat quantity in the pathogenesis of NAFLD and highlight the imperative for public health strategies focused on reducing saturated fat consumption to manage the growing burden of liver disease.

REFERENCES

1. Powell EE, Wong VW, Rinella M. Non-alcoholic fatty liver disease. *Lancet*. 2021; 397(10290): 2212-2224.
2. Samuel VT, Shulman GI. Nonalcoholic Fatty Liver Disease as a Nexus of Metabolic and Hepatic Diseases. *Cell Metab*. 2020; 27(1):22-41.
3. Mansouri A, Gattolliat CH, Asselah T. Mitochondrial Dysfunction and Signaling in Chronic Liver Diseases. *Gastroenterology*. 2021; 155(2): 629-649.
4. Aron-Wisnewsky J, Vigliotti C, Witjes J, Le P, Holleboom AG, Verheij J, et al. Gut microbiota and human NAFLD: disentangling microbial signatures from metabolic disorders. *Nat Rev Gastroenterol Hepatol.* 2020; 17(5):279-297.
5. Caussy C, Aubin A, Loomba R. The Relationship Between Type 2 Diabetes, NAFLD, and Cardiovascular Risk. *Curr Diab Rep*. 2021; 21(5):15.
6. Younossi ZM, Golabi P, Paik JM, Henry A, Van Dongen C, Henry L. The global epidemiology of nonalcoholic fatty liver disease (NAFLD) and nonalcoholic steatohepatitis (NASH): a systematic review. *Hepatology*. 2023; 77(4):1335-1347.
7. Younossi ZM, Golabi P, Paik JM, Henry A, Van Dongen C, Henry L. The global epidemiology of nonalcoholic fatty liver disease (NAFLD) and nonalcoholic steatohepatitis (NASH): a systematic review. *Hepatology*. 2023; 77(4):1335-47.
8. Li M, Xu C, Shi J, Lin J, Li X, Schouten JN, et al. Fatty acids promote fatty liver disease via the dysregulation of 3-mercaptopyruvate sulfurtransferase /hydrogen sulfide pathway. *Gut*. 2021; 70(11):2123-34.
9. Yoo JJ, Lee JS, Lee YB, Kim SG, Jang ES, Kim JH, et al. Relationship between dietary patterns and non-alcoholic fatty liver disease: A case-

control study. *Hepatobiliary Pancreat Dis Int.* 2022; 21(5):448-54.
10. Pandey A, Chawla S, Guchhait P. Type-2 diabetes: Current understanding and future perspectives. *IUBMB Life*. 2020; 72(11):2277-95.
11. Buzzetti E, Pinzani M, Tsochatzis EA. The multiple-hit pathogenesis of non-alcoholic fatty liver disease (NAFLD). *Metabolism*. 2020; 65:1038-48.
12. Garcia TS, Correa BR, de Araujo CM, Silva LD. The Impact of Dietary Fats on Liver Histology in Patients with Non-Alcoholic Fatty Liver Disease: A Systematic Review and Meta-Analysis. *Nutrients*. 2023; 15(2):438.
13. Silva J, Spaulding A, Deans L, Kamm DR, Lovering RM, Gao F, et al. Diet composition influences the metabolic benefits of short-term cycles of very low-calorie diets. *Physiol Rep.* 2021; 9(9):e14845.
14. Chen L, Li L, Chen J, Li L, Zheng Z, Ren J, et al. Oleic acid protects saturated fatty acid mediated lipotoxicity in hepatocytes and rat of non-alcoholic steatohepatitis. *Life Sci*. 2022; 203: 348-57.
15. Aron-Wisnewsky J, Vigliotti C, Witjes J, Le P, Holleboom AG, Verheij J, et al. Gut microbiota and human NAFLD: disentangling microbial signatures from metabolic disorders. *Nat Rev Gastroenterol Hepatol*. 2020; 17(5): 279-97.